



Shri Vile Parle Kelavani Mandal's

DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING

(Autonomous College Affiliated to the University of Mumbai)

NAAC Accredited with "A" Grade (CGPA : 3.18)



Department of Computer Science and Engineering (Data Science)
and Computer Engineering

Risk-Sensitive Portfolio Management with Multi-Agent Reinforcement Learning for Portfolio Optimization

Guide : Prof. Ruchika Patil

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Agenda

- Introduction
- Literature Review
- Gaps in Existing Work
- Gap Resolution
- Problem Definition and Objectives
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- Proposed Design
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Introduction

In today's volatile financial markets, portfolio optimization requires adaptive strategies that balance risk and return more effectively than traditional mean-variance or heuristic models. Leveraging Multi-Agent Reinforcement Learning (MARL), this project develops a smart, data-driven system where multiple agents collaborate to learn dynamic trading policies that respond to changing market conditions. Each agent specializes in goals such as maximizing returns or minimizing risk, and together they achieve robust, risk-adjusted performance. By integrating financial, technical, and macroeconomic data, the MARL framework enables continuous learning and adaptive portfolio allocation, outperforming static or single-agent approaches in real-world investment scenarios.

Literature Survey

Sr.no	Name of the Paper	Methods used	Results	Gaps
1	Portfolio Optimization of Indian Stock Market Using Reinforcement Learning and Explainable AI, (Rishabh Chorasiya, Shakti Kinger - 2025) -Technische Sicherheit	Single-agent DRL (PPO, A2C, DDPG, SAC) integrated with LIME explainability. Adds distributional critic for CVaR/DD; GNN for asset dependencies; online adaptive rebalancing	PPO best → Cumulative Return 34.04%, Sharpe 1.39, MaxDD -46.9%	• No tail-risk or drawdown-aware optimization • Ignores cross-asset correlations • Fails to adapt to changing market regimes
2	MARS – Meta-Adaptive Reinforcement Learning Framework for Risk-Aware Multi-Agent Portfolio Management, (Jiayi Chen-2025) - arXiv	Hierarchical MARL (Meta-Controller + Heterogeneous Agents with Safety Critics. GNN-based inter-asset communication; Top-k stock selection; heterogeneous agents for diverse data	DJI → Sharpe 2.84, MaxDD −5.39%; strong risk control	• Lacks modeling of asset inter-dependencies • Limited scalability across many assets • Agents not specialized by data type
3	Optimizing Trading Strategies in Quantitative Markets Using Multi-Agent RL (Zhang et al., 2023) - arXiv	Hybrid MADDPG with rule-based CPPI/TIPP overlays. Meta-controller learns multi-metric trade-offs; online regime-adaptive learning; specialized agents	TIPP-MADDPG → Annualized 9.68%, Sharpe 2.09, MaxDD -9.02%	• Static single-metric reward design • No dynamic regime adaptation • Homogeneous agent policies
4	RiskQ – Risk-Sensitive Multi-Agent RL Value Factorization (RiskQ, 2023)- arXiv	Distributional MARL with quantile regression and attention-based value mixing. Integrates regime-adaptive online updates; heterogeneous agent modules for financial signals	Theoretical results (SMAC benchmark); risk-aware coordination proven	No appropriate gaps

Sr.no	Name of the Paper	Methods used	Results	Gaps
5	Towards Multi-Agent RL-Driven Over-the-Counter Market Simulations, (Vadori et al., 2022) - arXiv	PPO-based multi-agent market simulator with RL-based calibration. Adds distributional risk critic; multi-objective reward; Top-k scalable selection mechanism	Realistic emergent LP/LT behaviors, no portfolio metrics	• No explicit tail-risk optimization • Single utility objective • High computational cost with many agents
6	Developing a Multi-Agent and Self-Adaptive Framework with Deep Reinforcement Learning for Dynamic Portfolio Risk Management, (Li et al. 2024) -arXiv	Proposed MASA framework with three agents — RL (TD3) for portfolio weights, solver-based agent for risk adjustment, and market-observer (DC/MLP/LSTM) for regime guidance. Uses a JSD-based action reward for diversity and joint optimization of return + risk.	Outperforms TD3-Profit and other baselines (stat. sig. via Wilcoxon test). MASA-MLP: AR 8.87%, MDD -31.78%, SR 0.27 (CSI 300).	• No CVaR/tail risk analysis. • Limited universe (Top-10 stocks) → selection bias.
7	Risk-Sensitive Deep Reinforcement Learning for Portfolio Optimization , (Wang & Liu - 2025) - JRFM	Transformer encoder to extract temporal/cross-asset features. Ensemble of DRL agents (DQN, PPO, A2C, DDPG) with adaptive agent-switching (ART-DRL).Volatility-sensitive reward function (penalizes downside; optimizes Sharpe/Sortino/Calmar).	ART-DRL produced best risk-adjusted metrics vs single agents. Reported strong Sharpe and Sortino improvements across market shocks	• No explicit futures roll / time-to-expiry or liquidity/market impact modeling. • Data proprietary → limited reproducibility.
8	Risk-Adjusted Deep Reinforcement Learning for Portfolio Optimization: A Multi-reward Approach, (Choudhary et al. 2025)	The model combines three DRL agents, each trained on a different reward function (Log Return for growth, DSR for risk-adjusted return, and MDD for risk reduction).	The model significantly outperformed standard benchmarks and market indices across four different stock exchanges.	• Fixed Portfolio: The model cannot handle a changing number of available stocks. • Offline Model and Blac box model

Sr.no	Name of the Paper	Methods used	Results	Gaps
9	MSPM: Modular & Scalable Multi-Agent RL for Portfolio Management , (Huang & Tanaka,2022)-Elsevier	Two-level modular design: Evolving Agent Modules (EAMs) per asset + Strategic Agent Module (SAM) for portfolio allocation.EAMs produce asset signals; SAM (PPO) aggregates EAM outputs to output portfolio weights.	Outperforms baselines (CRP, BAH, Adversarial PG) — 80% winning rate; ablation (disable EAM) causes big performance drop.	• Limited alt-data types tested; relies on FinSentS/EOD access. • No slippage/liquidity impact. • Scalability and compute cost when adding many EAMs not deeply evaluated.
10	Risk-Aware Distributed Multi-Agent Reinforcement Learning, (Al Maruf et al. ,2023) - arXiv	Uses CVaR for risk sensitivity .Proposes a distributed algorithm: CVaR QD-Learning Defines a CVaR Bellman operator and proves its convergence.	Bellman operator converges to the optimal value. Showed agents reach consensus on their value functions . Confirmed higher values for more risk-aware agents.	• Ignores malicious agents . • Limited to discrete spaces • Assumes deterministic costs in theory
11	Collaborative Multi-Agent Reinforcement Learning Model for Portfolio Management, (Hao et al. ,2025) - CIBDA	Cooperative multi-agent setup: one agent per asset; Agents trained with PPO; dynamic adjustment of U to balance individual vs group objectives. Discrete action grid for direction and allocation; normalized across assets.	Cooperative agents:profit $\approx 26,231$ vs non-coop and single-agent. Dynamic U and appropriate action discretization materially improve system returns and balance.	• Scalability to large portfolios not demonstrated • Discrete actions may limit fine allocation control vs continuous actions.
12	Risk-averse Reinforcement Learning for Portfolio Optimization , (B. Enkhsaikhan & O. Jo , 2024) - ICT Express	Formulated portfolio optimization as a risk-aware MDP with soft constraints. Used Actor–Critic (DDPG-style) reinforcement learning. Integrated Value-at-Risk (VaR) as a risk measure in the state. Monte Carlo sampling from BNN. Applied penalty function	$\sim 18\%$ lower risk with similar returns — better risk-adjusted performance than PPO/TRPO.	• Limited asset universe • Incomplete hyperparameter details (e.g., β, τ, χ not reported). • Parametric VaR assumption • Soft constraints only — no strict risk guarantees; relies on penalty tuning.

Gaps in Existing work

1) Tail-risk not considered

- Most MARL portfolio models optimize only mean returns; tail events like large drawdowns are ignored, leading to potential heavy losses.

2) Poor modeling of cross-asset dependencies

- Correlations, sector influence, and liquidity relationships between assets are often ignored. Agents act independently.

3) Single-objective optimization

- Only one metric (like return or Sharpe ratio) is optimized, ignoring the multi-metric nature of real-world portfolios.

4) Limited Feedback Integration

- Rewards often rely on single-step profit/loss.

Gap Resolution

1. Tail-risk optimization

Centralized distributional critic predicts the full return distribution. CVaR and max drawdown are integrated into the actor's reward, minimizing tail losses.

2. Cross-asset dependency modeling

Graph Neural Network (GNN) embeddings + inter-agent message passing capture correlations, sector influence, and liquidity links for coordinated decisions.

3. Multi-metric optimization

Rewards combine Return, Volatility, CVaR, max drawdown. A meta-agent dynamically learns trade-off weights to balance multiple objectives.

4. Optimizing feedback integration

Global Reward Module aggregates multi-metric feedback (return, volatility, drawdown, correlation) for stable learning.

Problem Definition

The primary problem is that traditional portfolio optimization methods are fundamentally inadequate for the dynamic and uncertain nature of financial markets, relying on static models that fail to capture real-time changes in risk and volatility. This results in portfolios that are brittle and sub-optimal, lacking the necessary adaptivity to effectively manage tail-risk events like sudden drawdowns while maximizing returns across different market regimes. Our project addresses this by proposing a Multi-Agent Reinforcement Learning (MARL) framework, where multiple cooperative agents learn a dynamic investment policy that explicitly optimizes a multi-objective, risk-adjusted reward, thereby creating a robust and intelligent system that adapts to market uncertainty in real time.

Objectives

- 1. Develop Multi-Agent RL Architecture**
- 2. Integrate Centralized Distributional Critic**
- 3. Implement Adaptive Rebalancing**
- 4. Optimize Top-k Stock Selection**
- 5. Design Realistic Execution & Feedback Loop**
- 6. Enable Explainability & Monitoring**

Dataset Description

Technical Indicators

SMA 20 – *Simple Moving Average (20-day)*

SMA 50 – *Simple Moving Average (50-day)*

RSI – *Relative Strength Index*

ADX – *Average Directional Index*

MACD – *Moving Average Convergence Divergence*

OBV – *On-Balance Volume*

A/D Line – *Accumulation/Distribution Line*

Bollinger Bands – *Bollinger Bands Indicator*

ATR – *Average True Range*

Dataset Description

External Factors (Macros)

Crude Oil WTI Future Historical Data

India VIX Historical Data

NASDAQ Composite Historical Data

S&P 500 Historical Data

USD_INR Historical Data

time	open	high	low	close	Volume
04-02-2013	2060	2084.3	2045.65	2064.6	375365
05-02-2013	2048.65	2109	2048.65	2094.7	443426
06-02-2013	2100	2118.3	2074	2074	479356
07-02-2013	2071.1	2081.45	2057	2063.8	179174
08-02-2013	2069	2094	2038.3	2046.85	432759
11-02-2013	2041.4	2062.25	2032.8	2044.75	310628
12-02-2013	2042.3	2084.9	2040.2	2076.15	245407
13-02-2013	2075	2082.2	2015	2018.8	223590
14-02-2013	2025	2026.9	1990.1	2002.6	210882
15-02-2013	1995.25	2010.15	1953.8	1970.35	285502
18-02-2013	1970.1	1980	1935	1946.6	244636
19-02-2013	1955	1996.2	1938.95	1988.05	214900
20-02-2013	1990.25	2002.4	1969.55	1982.7	320502
21-02-2013	1975	2002	1970.55	1983.55	265026
22-02-2013	1970	2019.9	1966.05	1993.65	219110
25-02-2013	2000	2020	1975.35	2009.55	281372
26-02-2013	1980	1998.9	1905	1921.6	364896
27-02-2013	1937	1975.8	1910	1964.1	440033
28-02-2013	1979.9	1993.8	1955.4	1973	586457
01-03-2013	1973	2042.2	1947	2028.3	494169
04-03-2013	2024.7	2025.45	1966.05	1989.2	499009
05-03-2013	1978	1998.9	1950.7	1973.4	785020
06-03-2013	1976	1986.1	1966	1973.3	404202

NSE_BAJAJ_AUTO

Date	Price	Open	High	Low	Vol.	Change %
01-12-2014	2,053.40	2,065.80	2,065.80	2,049.60		-0.69%
02-12-2014	2,066.60	2,053.80	2,068.80	2,053.80		0.64%
03-12-2014	2,074.30	2,067.40	2,076.30	2,066.70		0.37%
04-12-2014	2,071.90	2,073.60	2,077.30	2,062.30		-0.12%
05-12-2014	2,075.40	2,072.80	2,079.50	2,070.80		0.17%
08-12-2014	2,060.30	2,074.80	2,075.80	2,054.30		-0.73%
09-12-2014	2,059.80	2,056.60	2,060.60	2,034.20		-0.02%
10-12-2014	2,026.10	2,058.90	2,058.90	2,024.30		-1.64%
11-12-2014	2,035.30	2,027.90	2,055.50	2,027.90		0.45%
12-12-2014	2,002.30	2,030.40	2,032.20	2,002.30		-1.62%
15-12-2014	1,989.60	2,005.00	2,018.70	1,982.30		-0.63%
16-12-2014	1,972.70	1,986.70	2,016.90	1,972.60		-0.85%
17-12-2014	2,012.90	1,973.80	2,016.80	1,973.80		2.04%
18-12-2014	2,061.20	2,019.00	2,061.20	2,019.00		2.40%
19-12-2014	2,070.70	2,061.00	2,077.80	2,061.00		0.46%
22-12-2014	2,078.50	2,069.30	2,078.80	2,069.30		0.38%
23-12-2014	2,082.20	2,081.50	2,086.70	2,079.80		0.18%
24-12-2014	2,081.90	2,083.20	2,087.60	2,081.90		-0.01%
26-12-2014	2,088.80	2,084.30	2,092.70	2,084.30		0.33%
29-12-2014	2,090.60	2,087.60	2,093.60	2,085.80		0.09%
30-12-2014	2,080.30	2,088.50	2,088.50	2,079.50		-0.49%
31-12-2014	2,058.90	2,082.10	2,085.60	2,057.90		-1.03%
02-01-2015	2,058.20	2,058.90	2,072.40	2,046.00		-0.03%

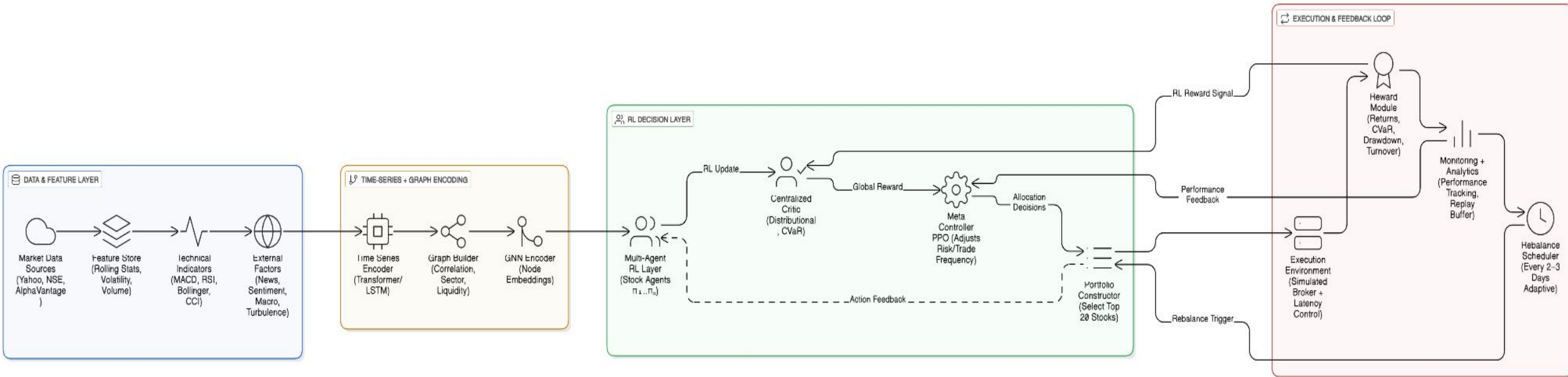
S&P 500 Historical Data

Dataset Description

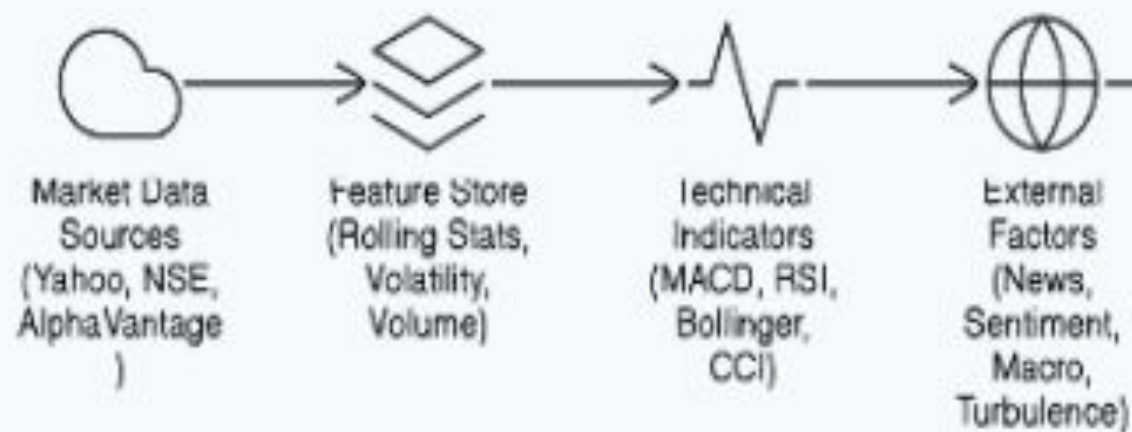
- **Yahoo Finance API:** Allows fetching real-time stock prices, historical data, and market summaries. Though it is unofficial and has some limitations for high-frequency updates, it is widely used for educational and prototyping purposes.
- **Alpha Vantage API:** Provides official access to real-time and intraday stock prices, technical indicators, and historical data with a well-defined request structure. It also supports multiple asset classes like equities, forex, and cryptocurrencies.
- In practice, the dataset that was used for **backtesting** (historical prices, indicators, and technical features) can be extended with **real-time data** fetched from these APIs. This allows the strategy to **simulate live trading conditions**, monitor the system's performance on current market movements, and refine trading decisions based on **up-to-date market information**.

Essentially, these APIs **bridge the gap between historical simulations and live deployment**, ensuring that the model not only performs well on past data but also adapts to ongoing market dynamics.

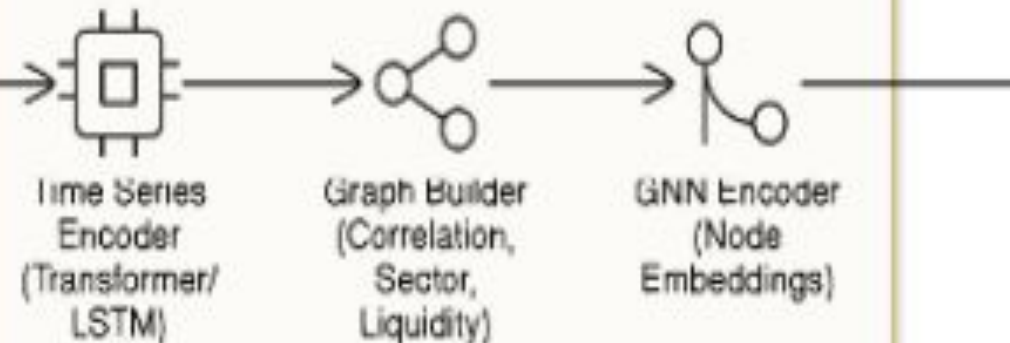
Proposed Design

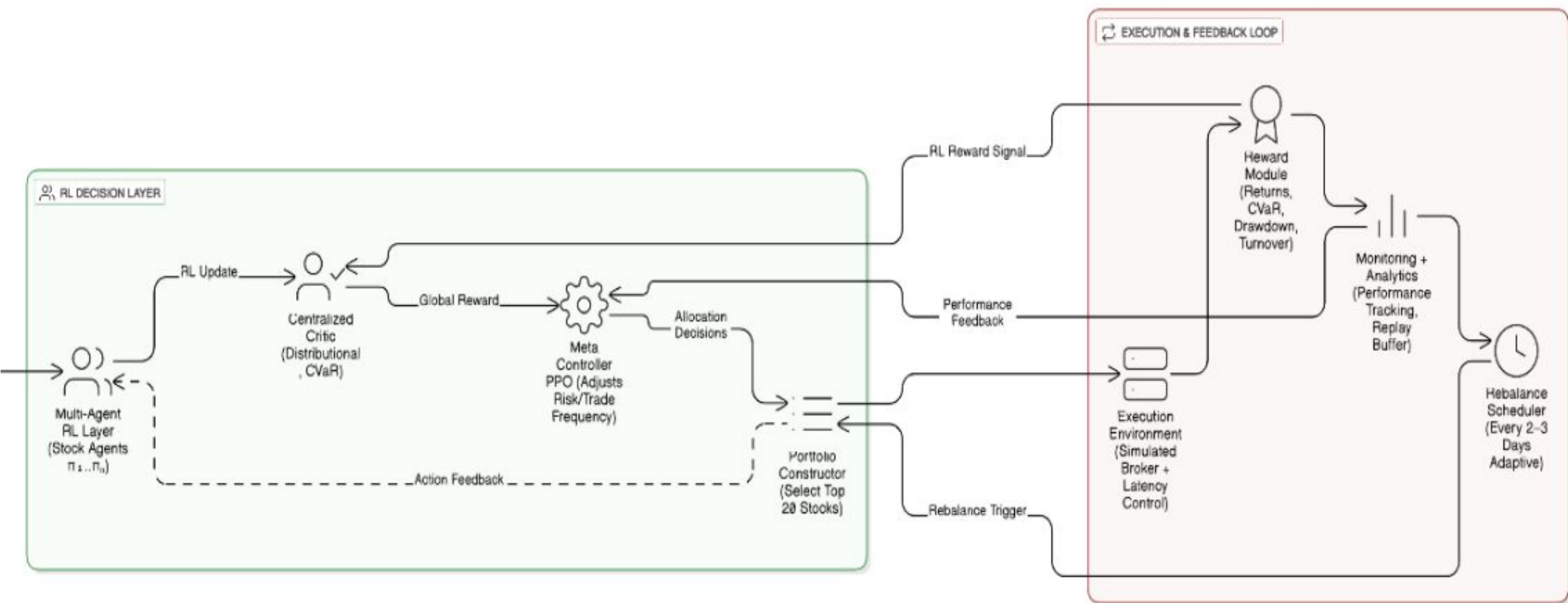


DATA & FEATURE LAYER

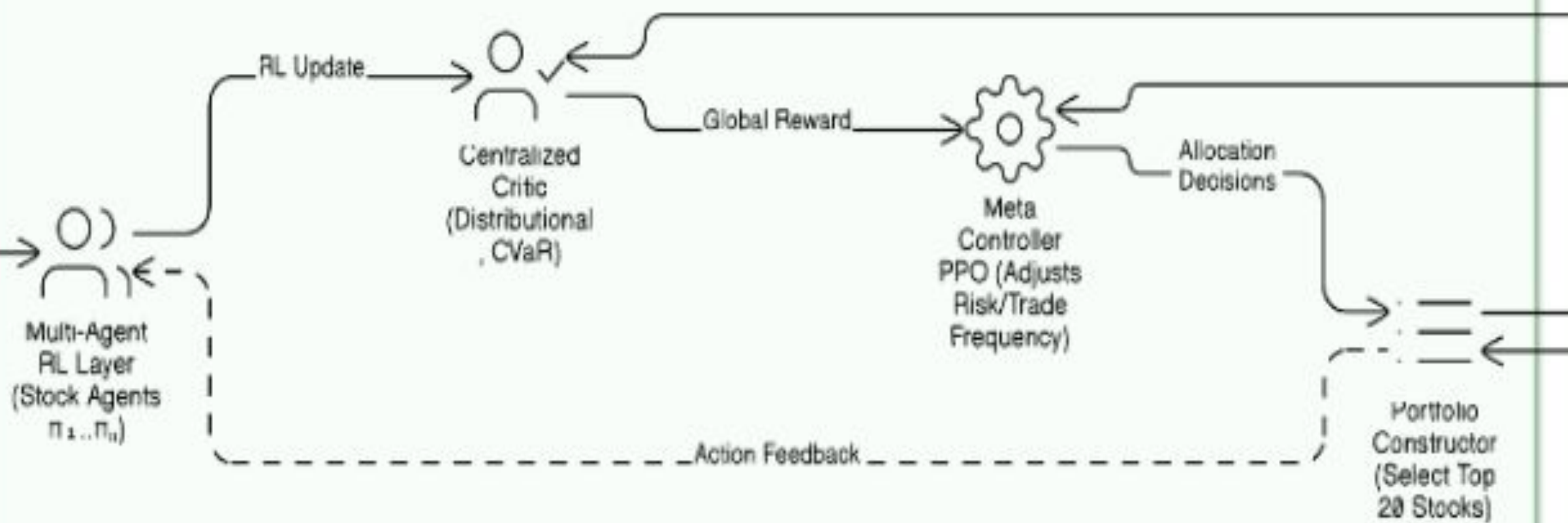


TIME-SERIES + GRAPH ENCODING





RL DECISION LAYER



Mathematical Formulation for the Architecture

1. Actor-update (Multi-agent RL)

A single stock agent's policy (μ_i) receives a **Node Embedding (zi)** from a GNN as input. It then updates its policy to maximize the **risk-adjusted value ($\psi\alpha$)** provided by the critic. This means the agent learns to choose actions that not only lead to high returns but also result in a favorable **tail-risk profile** (low CVaR). This is the core mechanism that makes the agents risk-sensitive.

$$\nabla_{\theta_i} J(\theta_i) \approx \mathbb{E}_{\mathbf{o}} [\nabla_{\mathbf{a}_i} \psi_{\alpha} [\mathbf{Z}(\mathbf{o}, \mathbf{a})] \cdot \nabla_{\theta_i} \mu_i(\mathbf{o})]_{\mathbf{a}=\mu(\mathbf{o})}$$

Symbols:

- $\nabla_{\theta_i} J(\theta_i)$: The **gradient of the objective function** with respect to the actor's parameters.
- θ_i : **Parameters** of the i -th actor network.
- $\psi_{\alpha}[\cdot]$: A **risk measure** (e.g., CVaR) applied to the return distribution.
- $\mathbf{Z}(\mathbf{o}, \mathbf{a})$: The **return distribution** predicted by the critic for a given state-action pair.
- $\nabla_{\mathbf{a}_i} \psi_{\alpha}[\cdot]$: The **gradient of the risk measure** with respect to the action.
- $\nabla_{\theta_i} \mu_i(\mathbf{o})$: The **gradient of the actor's policy**.

2. Critic Update (Centralised critic)

A Centralized Critic receives the **Reward (r)** from the Execution Layer. It compares this actual reward to its previous prediction of the return distribution. The equation represents this learning process: the critic uses the **Quantile Regression Loss** to adjust its internal parameters. This ensures its predicted return distribution for a given market state accurately reflects real-world outcomes, including **tail risks**. The updated critic then serves as the "teacher" for the individual agents.

$$\mathcal{L}_{\text{Critic}} = \mathbb{E}_{\mathbf{o}, \mathbf{a}, r, \mathbf{o}'} \left[\sum_{k=1}^N \rho_{\tau_k}^{\kappa}(\delta_k) \right]$$

Symbols:

- $\mathcal{L}_{\text{Critic}}$: Critic network's **loss function**.
- $\mathbb{E}$: **Expected value** over a batch of experiences.
- $\mathbf{o}, \mathbf{a}, r, \mathbf{o}'$: A single **transition tuple** (state, action, reward, next state).
- $\rho_{\tau_k}^{\kappa}(\delta_k)$: The **Quantile Huber Loss**.
- δ_k : The **Temporal Difference (TD) error** for the k -th quantile.
- N : The **number of quantiles** used to approximate the return distribution.

3.CVaR integration (Centralised critic)

The **Distributional Critic** in the architecture predicts the full **Return Distribution (Z)** of the portfolio. From this distribution, the CVaR is calculated. The metric is then used as a direct penalty term in the overall **Reward Function**, forcing the entire system to avoid scenarios where extreme losses are likely. It is a critical component of the **Risk Management Overlay** that the architecture provides.

$$\text{CVaR}_\alpha(\mathbf{Z}) = \mathbb{E}[Z | Z \leq \text{VaR}_\alpha(Z)]$$

Symbols:

- $\text{CVaR}_\alpha(\mathbf{Z})$: The **Conditional Value at Risk** at the α confidence level.
- $\mathbf{Z}$: The **random variable** representing the portfolio's return.
- $\mathbb{E}[Z | Z \leq \text{VaR}_\alpha(Z)]$: The **expected value** of returns in the worst $\alpha\%$ of outcomes.
- $\text{VaR}_\alpha(Z)$: The **Value at Risk**, or the threshold value below which returns fall $\alpha\%$ of the time.

4. Reward Function(Multi-metric trade off)

The equation represents a **multi-objective reward function** combining return, volatility, CVaR, and maximum drawdown into a unified signal. It encourages the agent to **maximize returns** while **minimizing overall risk and portfolio instability**, with adaptive weights allowing the system to dynamically balance risk–return trade-offs under changing market conditions.

$$R = w_{\text{ret}} \cdot \text{ret} + w_{\text{vol}} \cdot (-\text{vol}) + w_{\text{cvar}} \cdot (-\text{cvar}) + w_{\text{mdd}} \cdot (-\text{mdd})$$

RESULTS

Smoke Test Run: The smoke test was conducted to verify the seamless integration of all core components — ensuring that the data pipeline, encoders, graph modules, and agent interactions function coherently as a unified system, confirming the setup's readiness for full-scale training.

```
(.venv) dhruvs@LAPTOP-5IR4COF4:/mnt/d/fyp/proj_implementation$ python smoke_test.py
Building windows for 5 stocks...
Building encoder...
✓ Encoder output shape: torch.Size([5, 64])
(.venv) dhruvs@LAPTOP-5IR4COF4:/mnt/d/fyp/proj_implementation$ python smoke_test.py
Building windows for 32 stocks...
Building encoder...
✓ Encoder output shape: torch.Size([32, 64])
```


ACTOR CRITIC STOCK SELECTION DURING TRAINING

date	nav	reward	selected
2015-09-10T00:00:00.000000000	1	0	
2015-09-11T00:00:00.000000000	1	0	
2015-09-14T00:00:00.000000000	1	0	NSE_ONGC,NSE_SOLARINDS,NSE_CIPLA,NSE_HINDUNILVR,NSE_PIDILITIND,NSE_TRENT,NSE_ASIANPAINT
2015-09-15T00:00:00.000000000	1	0	NSE_ONGC,NSE_SOLARINDS,NSE_CIPLA,NSE_HINDUNILVR,NSE_TRENT,NSE_PIDILITIND,NSE_ASIANPAINT,NSE_INDHOTEL
2015-09-16T00:00:00.000000000	1	0	NSE_ONGC,NSE_CIPLA,NSE_SOLARINDS,NSE_ABB,NSE_HINDUNILVR,NSE_PIDILITIND,NSE_TRENT,NSE_INDHOTEL,NSE_TCS
2015-09-18T00:00:00.000000000	1	0	NSE_ONGC,NSE_CIPLA,NSE_SOLARINDS,NSE_ABB,NSE_HINDUNILVR,NSE_PIDILITIND,NSE_TRENT,NSE_INDHOTEL
2015-09-21T00:00:00.000000000	1	0	NSE_ONGC,NSE_SOLARINDS,NSE_CIPLA,NSE_ABB,NSE_HINDUNILVR,NSE_PIDILITIND,NSE_TRENT,NSE_BEL
2015-09-22T00:00:00.000000000	1	0	NSE_ONGC,NSE_ABB,NSE_CIPLA,NSE_SOLARINDS,NSE_HINDUNILVR,NSE_PIDILITIND,NSE_TRENT,NSE_BEL
2015-09-23T00:00:00.000000000	1	0	NSE_ONGC,NSE_ABB,NSE_CIPLA,NSE_SOLARINDS,NSE_PIDILITIND,NSE_TRENT,NSE_HINDUNILVR
2015-09-24T00:00:00.000000000	1	0	NSE_ONGC,NSE_ABB,NSE_SOLARINDS,NSE_CIPLA,NSE_PIDILITIND,NSE_TRENT,NSE_ASIANPAINT,NSE_HINDUNILVR
2015-09-28T00:00:00.000000000	1	0	NSE_ONGC,NSE_SOLARINDS,NSE_CIPLA,NSE_ABB,NSE_PIDILITIND,NSE_TRENT,NSE_ASIANPAINT,NSE_HINDUNILVR
2015-09-29T00:00:00.000000000	1	0	NSE_SOLARINDS,NSE_ABB,NSE_ONGC,NSE_CIPLA,NSE_PIDILITIND,NSE_TRENT,NSE_ASIANPAINT,NSE_BEL
2015-09-30T00:00:00.000000000	1	0	NSE_SOLARINDS,NSE_CIPLA,NSE_ONGC,NSE_ABB,NSE_PIDILITIND,NSE_AMBUJACEM,NSE_TRENT,NSE_ASIANPAINT,NSE_BEL
2015-10-01T00:00:00.000000000	1	0	NSE_CIPLA,NSE_ONGC,NSE_HDFCBANK,NSE_ABB,NSE_AMBUJACEM,NSE_PIDILITIND,NSE_SOLARINDS,NSE_TRENT,NSE_BEL
2015-10-05T00:00:00.000000000	1	0	NSE_ONGC,NSE_AMBUJACEM,NSE_ABB,NSE_PIDILITIND,NSE_CIPLA,NSE_HDFCBANK,NSE_SOLARINDS,NSE_TRENT,NSE_BEL
2015-10-06T00:00:00.000000000	1	0	NSE_AMBUJACEM,NSE_ABB,NSE_CIPLA,NSE_HDFCBANK,NSE_ONGC,NSE_PIDILITIND,NSE_SOLARINDS,NSE_TRENT
2015-10-07T00:00:00.000000000	1	0	NSE_ABB,NSE_AMBUJACEM,NSE_CIPLA,NSE_ONGC,NSE_PIDILITIND,NSE_HDFCBANK,NSE_SOLARINDS,NSE_TRENT
2015-10-08T00:00:00.000000000	1	0	NSE_ABB,NSE_CIPLA,NSE_AMBUJACEM,NSE_PIDILITIND,NSE_HDFCBANK,NSE_ONGC,NSE_GODREJCP,NSE_SOLARINDS,NSE_TRENT
2015-10-09T00:00:00.000000000	1	0	NSE_ABB,NSE_HDFCBANK,NSE_CIPLA,NSE_PIDILITIND,NSE_ONGC,NSE_GODREJCP,NSE_AMBUJACEM,NSE_SOLARINDS,NSE_TRENT
2015-10-12T00:00:00.000000000	1	0	NSE_HDFCBANK,NSE_CIPLA,NSE_ABB,NSE_GODREJCP,NSE_PIDILITIND,NSE_ONGC,NSE_AMBUJACEM,NSE_SOLARINDS,NSE_TRENT
2015-10-13T00:00:00.000000000	1	0	NSE_CIPLA,NSE_GODREJCP,NSE_ABB,NSE_PIDILITIND,NSE_HDFCBANK,NSE_ONGC,NSE_AMBUJACEM,NSE_HINDUNILVR,NSE_TRENT,NSE_SOLARINDS
2015-10-14T00:00:00.000000000	1	0	NSE_GODREJCP,NSE_CIPLA,NSE_ABB,NSE_PIDILITIND,NSE_HDFCBANK,NSE_TRENT,NSE_AMBUJACEM,NSE_SOLARINDS,NSE_ONGC
2015-10-15T00:00:00.000000000	1	0	NSE_GODREJCP,NSE_CIPLA,NSE_ABB,NSE_HDFCBANK,NSE_PIDILITIND,NSE_SOLARINDS,NSE_TRENT,NSE_ONGC
2015-10-16T00:00:00.000000000	1	0	NSE_CIPLA,NSE_ABB,NSE_GODREJCP,NSE_ONGC,NSE_SOLARINDS,NSE_HDFCBANK,NSE_TRENT,NSE_BEL
2015-10-19T00:00:00.000000000	1	0	NSE_ABB,NSE_GODREJCP,NSE_CIPLA,NSE_HDFCBANK,NSE_ONGC,NSE_SOLARINDS,NSE_TRENT,NSE_INDHOTEL
2015-10-20T00:00:00.000000000	1	0	NSE_GODREJCP,NSE_ABB,NSE_SOLARINDS,NSE_ONGC,NSE_HDFCBANK,NSE_TRENT,NSE_INDHOTEL,NSE_CIPLA
2015-10-21T00:00:00.000000000	1	0	NSE_GODREJCP,NSE_ABB,NSE_SOLARINDS,NSE_CIPLA,NSE_ONGC,NSE_TRENT,NSE_INDHOTEL
2015-10-23T00:00:00.000000000	1	0	NSE_SOLARINDS,NSE_GODREJCP,NSE_CIPLA,NSE_PIDILITIND,NSE_ABB,NSE_ONGC,NSE_TRENT,NSE_INDHOTEL
2015-10-26T00:00:00.000000000	1	0	NSE_GODREJCP,NSE_SOLARINDS,NSE_CIPLA,NSE_TRENT,NSE_PIDILITIND,NSE_ONGC,NSE_INDHOTEL,NSE_BEL,NSE_ABB

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